

SOUTH WEYMOUTH NAS, MA, USA

WMO: 725097

Lat: 42.150N

Lon: 70.933W

Elev: 161

StdP: 14.61

Time Zone: -5.00 (NAE)

Period: 82-93

WBAN: 14790

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.9	10.4	-13.9	2.6	9.7	-8.1	3.6	14.0	20.0	31.5	18.3	28.5	6.4	300	0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	91.2	73.8	87.7	72.3	84.7	70.7	76.9	86.8	74.9	83.8	73.1	81.4	8.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	127.9	81.9	72.2	119.7	79.4	70.3	112.2	77.6	40.4	87.2	38.4	84.2	36.7	81.4	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	16.5	14.5	DB	-2.0	95.5	3.7	2.5	-4.6	97.3	-6.8	98.7	-8.9	100.1	-11.5	101.9	(4)
(5)			WB	-3.5	80.0	3.5	1.7	-6.0	81.2	-8.1	82.2	-10.0	83.1	-12.6	84.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	28.2	31.0	38.1	47.6	58.0	66.5	72.3	70.9	63.1	53.7	44.2	34.7		
	DBStd	16.97	9.57	8.87	8.71	7.46	8.02	6.88	5.76	6.30	7.22	7.66	8.18	9.45		
	HDD50	2492	677	534	383	133	18	0	0	0	2	49	215	481		
	HDD65	5832	1142	953	835	525	246	62	10	18	119	359	625	938		
	CDD50	2784	1	2	13	60	267	495	691	648	396	163	40	8		
	CDD65	646	0	0	0	1	30	107	236	201	63	8	0	0		
	CDH74	5734	0	0	3	57	397	1092	2046	1628	468	39	4	0		
	CDH80	1994	0	0	0	19	139	394	749	555	136	2	0	0		
(14) Wind	WSAvg	6.2	6.4	7.1	7.3	7.1	6.4	5.9	5.2	5.1	5.0	5.7	6.4	6.5		
(15) Precipitation	PrecAvg	43.8	4.0	3.9	3.8	3.5	3.2	2.9	2.5	3.4	3.5	3.7	4.3	4.3		
	PrecMax	59.5	10.7	7.6	9.8	7.6	6.7	10.4	4.9	8.0	8.3	8.9	9.6	9.5		
	PrecMin	25.4	0.5	0.6	0.7	1.1	0.5	0.6	0.7	0.6	0.5	1.2	0.7	1.1		
	PrecStd	7.7	2.2	1.6	2.0	1.5	1.6	2.0	1.2	2.0	2.2	1.6	2.2	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	60.8	71.5	81.6	90.6	95.1	95.8	93.8	88.9	78.7	72.6	62.4		
		MCWB	51.8	51.6	55.2	62.3	69.1	73.5	76.4	78.8	75.7	66.1	61.4	57.9		
	2%	DB	50.8	53.7	61.6	71.4	83.6	89.1	91.0	89.7	84.2	73.8	66.2	57.6		
		MCWB	46.7	48.5	52.1	57.7	66.9	71.3	74.4	74.5	72.0	64.6	59.0	53.8		
	5%	DB	45.8	49.2	55.9	65.7	77.9	84.2	87.3	85.9	79.3	70.2	61.8	54.0		
		MCWB	41.4	44.6	48.8	53.8	64.1	68.6	71.9	72.6	68.4	62.5	55.9	51.0		
	10%	DB	42.0	44.9	51.5	60.0	72.7	80.4	84.1	82.6	75.2	66.7	58.1	49.1		
		MCWB	38.1	40.1	45.1	51.1	60.2	66.3	70.8	70.7	66.4	59.5	53.4	44.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	54.3	58.3	64.1	73.5	75.7	79.2	80.2	76.9	70.3	64.7	59.3		
		MCDB	56.6	57.4	65.6	79.0	88.1	87.7	89.6	92.1	87.3	73.3	68.6	62.2		
	2%	WB	47.2	49.1	53.6	59.7	68.9	73.3	76.8	77.2	73.3	67.1	60.2	54.9		
		MCDB	50.7	52.9	60.6	68.4	79.3	86.8	87.4	86.1	81.0	71.8	64.8	57.1		
	5%	WB	42.2	45.5	49.9	56.1	65.5	71.0	74.7	75.0	71.0	64.0	56.9	50.8		
		MCDB	45.2	49.6	55.4	63.0	75.1	82.1	83.8	82.2	77.4	68.8	60.8	53.9		
	10%	WB	38.3	40.5	45.5	52.5	62.3	68.3	72.8	73.0	68.1	61.0	53.5	45.6		
		MCDB	41.5	44.5	51.0	59.0	70.6	77.3	80.4	80.1	73.2	65.5	57.6	48.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	17.2	17.7	17.9	20.6	20.9	19.8	18.6	19.7	19.3	17.1	16.3		
		MCDBR	20.1	21.4	25.7	27.8	29.0	27.6	24.4	22.6	22.4	22.0	23.6	20.0		
		MCWBR	18.1	18.1	18.7	16.6	16.3	14.8	12.3	10.6	12.9	14.5	19.0	19.8		
	5% WB	MCDBR	19.6	20.9	23.2	24.7	25.2	24.3	21.3	19.0	19.6	18.1	20.1	20.1		
		MCWBR	19.2	18.9	19.0	17.1	16.0	14.8	12.0	10.3	13.5	13.3	18.1	20.4		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.303	0.326	0.368	0.405	0.452	0.473	0.432	0.373	0.335	0.310	0.296		
	taud		2.510	2.498	2.487	2.394	2.331	2.233	2.182	2.283	2.429	2.542	2.573	2.569		
	Ebn at Noon		273	288	293	286	277	263	255	263	273	273	266	264		
	Edh at Noon		23	27	30	36	40	44	46	40	32	26	21	20		
(44) All-Sky Solar Radiation	RadAvg		562	839	1159	1493	1665	1800	1884	1694	1365	913	618	467		
	RadStd		39	75	82	160	177	170	111	78	94	79	47	50		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (69 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+77

Nomenclature: See separate page